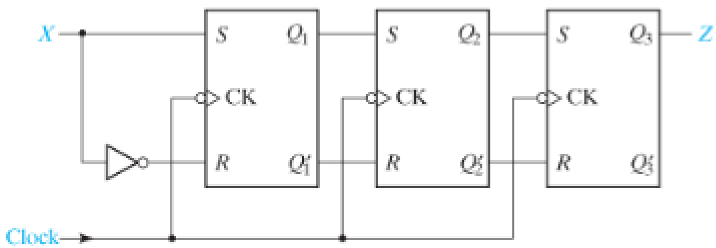
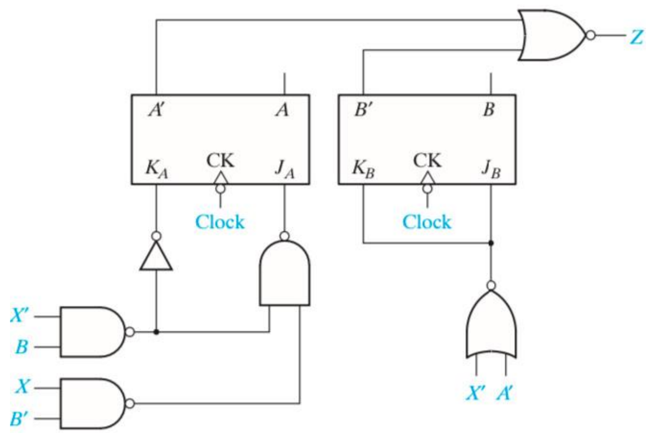
ECE 3441 Homework **Unit 13** **Name/ID#**________________________

13.2 – 13.5, 13.13, 13.15, 13.21, 13.27, 13.30

13.2 Construct a transition graph for the shift register shown. (X is the input, and Z is the output.) Is this a Mealy or Moore machine?



13.3 (a) For the following sequential circuit, find the next-state equation or map for each flip-flop. (Is this a Mealy or Moore machine?) Using these next-state equations or maps, construct a transition table and graph for the circuit.

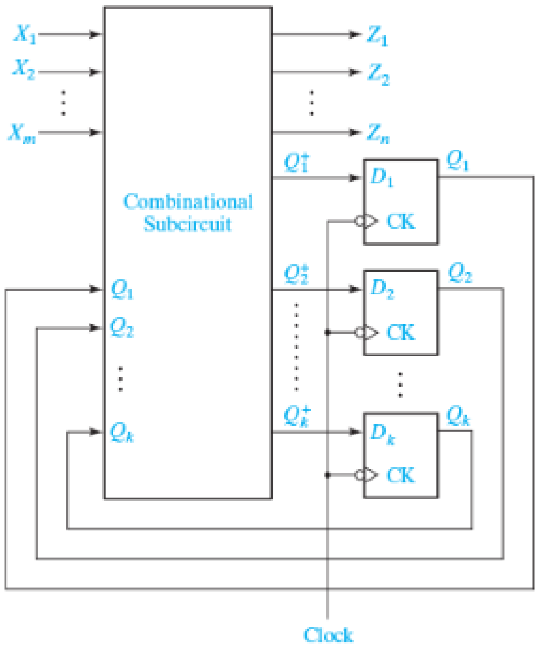


(b) What is the output sequence when the input sequence is $X = 01100$?

(c) Draw a timing diagram for the input sequence in (b). Show the clock, X, A, B, and Z. Assume that the input changes between falling and rising clock edges.

13.4 A sequential circuit has the form shown in Figure 13-17, with

$$D_1 = Q_2Q_3' \quad D_3 = Q_2' + X$$
$$D_2 = Q_3 \quad Z = XQ_2' + X'Q_2$$



a. Construct a transition table and state graph for the circuit. (Is this a Mealy or Moore machine?)

b. Draw a timing diagram for the circuit showing the clock, X , Q_1 , Q_2 , Q_3 and Z . Use the input sequence $X = 01011$. Change X between clock edges so that we can see false outputs, and indicate any false outputs on the diagram.

c. Compare the output sequence obtained from the timing diagram with that from the transition graph.

d. At what time with respect to the clock should the input be changed in order to eliminate the false output(s)?

13.5 Below is a transition table with the outputs missing. The output should be $Z = X'B' + XB$.

	$A^+B^+C^+$		
ABC	$X=0$	$X=1$	
000	011	010	
001	000	100	
010	100	100	
011	010	000	
100	100	001	

a. Is this a Mealy machine or Moore machine?

b. Fill in the outputs on the transition table.

c. Give the transition graph.

d. For an input sequence of $X = 10101$, give a timing diagram for the clock, X , A , B , C , and Z . State changes occur on the rising clock edge. What is the correct output sequence for Z ? Change X between rising and falling clock edges so that we can see false outputs, and indicate any false outputs on the diagram.

13.13 A sequential circuit has one input X , one output Z , and three flip-flops Q_1 , Q_2 and Q_3 . The transition and output tables for the circuit follow:

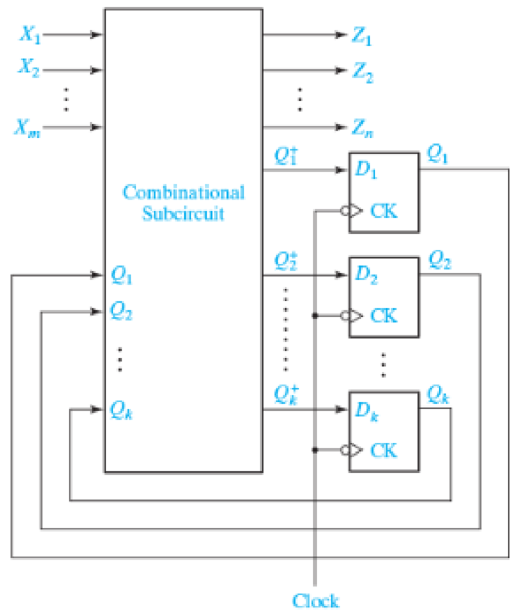
	Next State		Output (Z)	
$X=0$	$X=1$	$X=0$	$X=1$	
000	100	101	1	0
001	100	101	0	1
010	000	000	1	0
011	000	000	0	1
100	111	110	1	0
101	110	110	0	1
110	011	010	1	0
111	011	011	0	1

a. Construct a timing chart for the input sequence $X = 0101$ and initial state $Q_1Q_2Q_3 = 000$. Identify any false outputs. (Assume that the flip-flops are rising-edge triggered and that the input changes midway between the rising and falling edges of the clock.)

b. List the output values produced by the input sequence.

13.15 A sequential circuit has the form shown in Figure 13-17 with

$$D_1 = Q_2Q_3' + XQ_1' \quad D_3 = Q_2' + X$$
$$D_2 = Q_3 + X'Q_2 \qquad Z = XQ_2' + X'Q_2$$



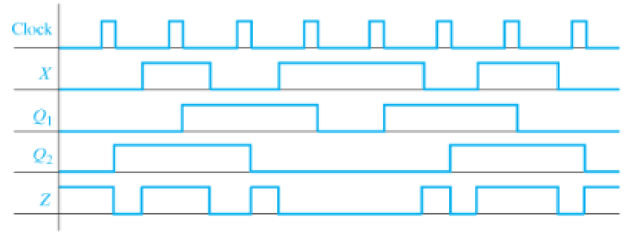
a. Construct a state table and state graph for the circuit.

b. Draw a timing diagram for the circuit showing the clock, X, Q_1, Q_2, Q_3, and Z. Use the input sequence $X = 01011$ and assume that X changes midway between falling and rising clock edges. Indicate any false outputs on the diagram.

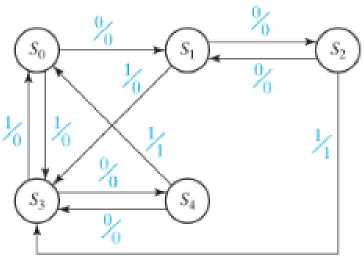
c. Compare the output sequence obtained from the timing diagram with that from the state graph.

d. At what time with respect to the clock should the input be changed in order to eliminate the false output(s)?

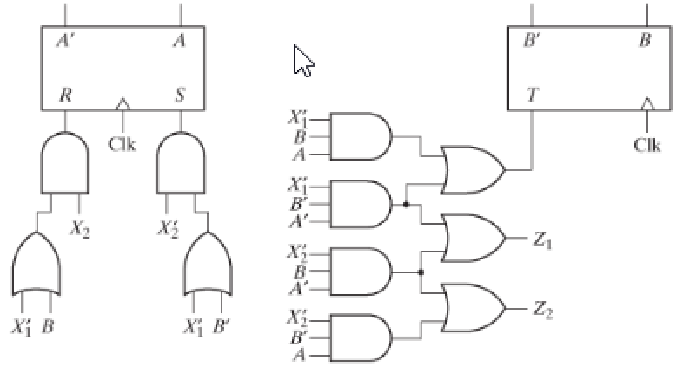
13.21 A Mealy sequential circuit has one input, one output, and two flip-flops. A timing diagram for the circuit follows. Construct a transition table and state graph for the circuit.



13.27 For the following state graph, construct a transition table. Then, give the timing diagram for the input sequence $X = 101001$. Assume X changes midway between the falling and rising edges of the clock, and that the flip-flops are falling-edge triggered. What is the correct output sequence?



13.30 a. For the following sequential circuit, write the next-state equations for flip-flops A and B.



b. Using these equations, find the transition table and draw the state graph.